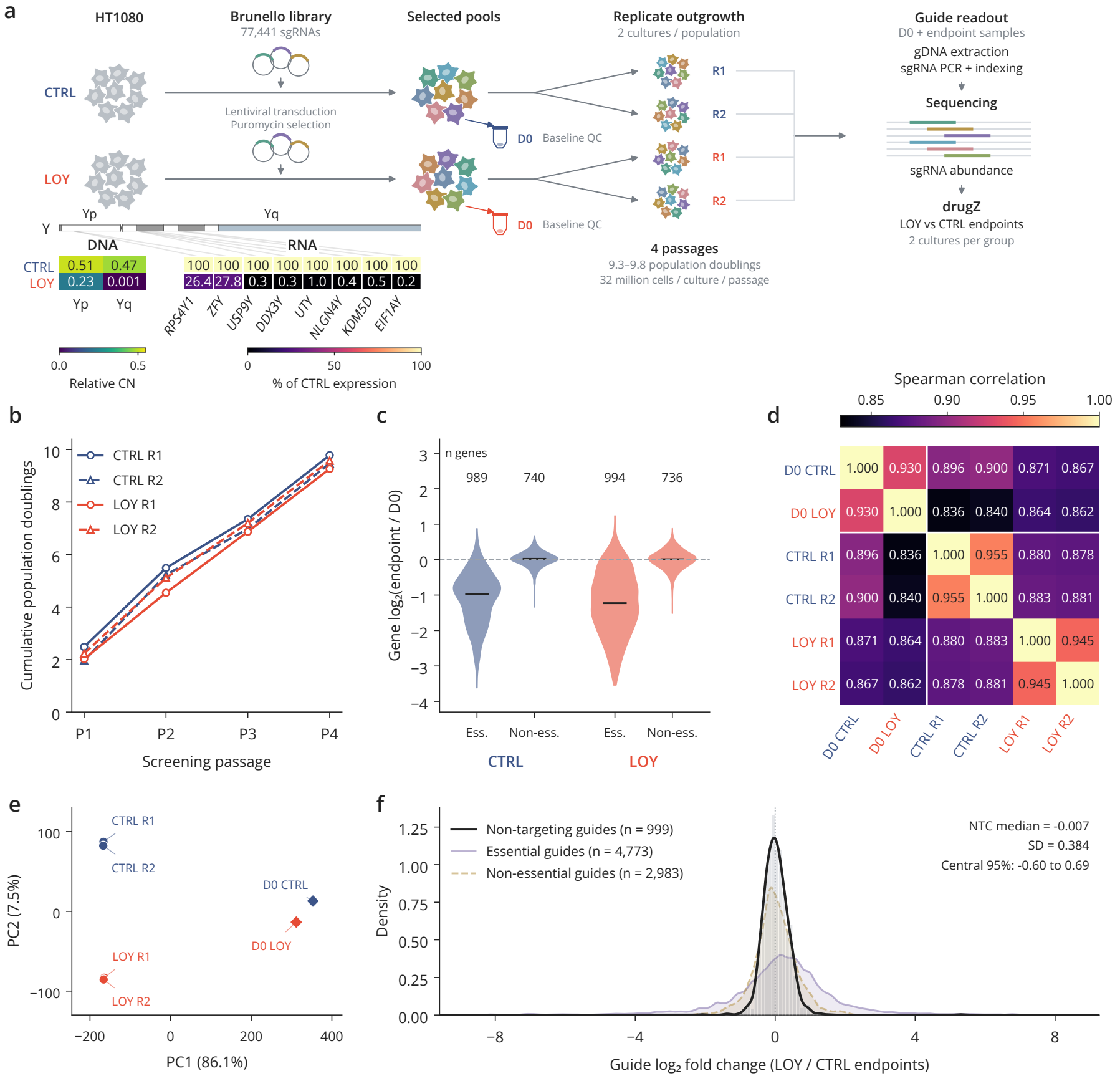

a
HT1080
Brunello library
77,441 sgRNAs
Selected pools
Replicate outgrowth
2 cultures / population
Guide readout
D0 + endpoint samples
gDNA extraction
sgRNA PCR + indexing
Sequencing
sgRNA abundance
drugZ
LOY vs CTRL endpoints
2 cultures per group
CTRL
LOY
Lentiviral transduction
Puromycin selection
D0 Baseline QC
D0 Baseline QC
R1
R2
R1
R2
4 passages
9.3–9.8 population doublings
32 million cells / culture / passage
Y
Yp
Yq
DNA
RNA
CTRL 0.51 0.47
LOY 0.23 0.001
Yp Yq
100 100 100 100 100 100 100 100
26.4 27.8 0.3 0.3 1.0 0.4 0.5 0.2
RPS4Y1 ZFY USP9Y DDX3Y UTY NLGN4Y KDM5D EIF1AY
0.0 0.5
Relative CN
0 50 100
% of CTRL expression
b
Cumulative population doublings
CTRL R1
CTRL R2
LOY R1
LOY R2
P1 P2 P3 P4
Screening passage
c
n genes
989 740 994 736
Gene log2(endpoint / D0)
Ess. Non-ess. Ess. Non-ess.
CTRL LOY
d
Spearman correlation
0.85 0.90 0.95 1.00
D0 CTRL 1.000 0.930 0.896 0.900 0.871 0.867
D0 LOY 0.930 1.000 0.836 0.840 0.864 0.862
CTRL R1 0.896 0.836 1.000 0.955 0.880 0.878
CTRL R2 0.900 0.840 0.955 1.000 0.883 0.881
LOY R1 0.871 0.864 0.880 0.883 1.000 0.945
LOY R2 0.867 0.862 0.878 0.881 0.945 1.000
D0 CTRL D0 LOY CTRL R1 CTRL R2 LOY R1 LOY R2
e
CTRL R1
CTRL R2
D0 CTRL
D0 LOY
LOY R1
LOY R2
PC2 (7.5%)
PC1 (86.1%)
f
Non-targeting guides (n = 999)
Essential guides (n = 4,773)
Non-essential guides (n = 2,983)
NTC median = -0.007
SD = 0.384
Central 95%: -0.60 to 0.69
Density
Guide log2 fold change (LOY / CTRL endpoints)